inclass-L33

Thursday, October 8, 2026 12:34 PM

CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

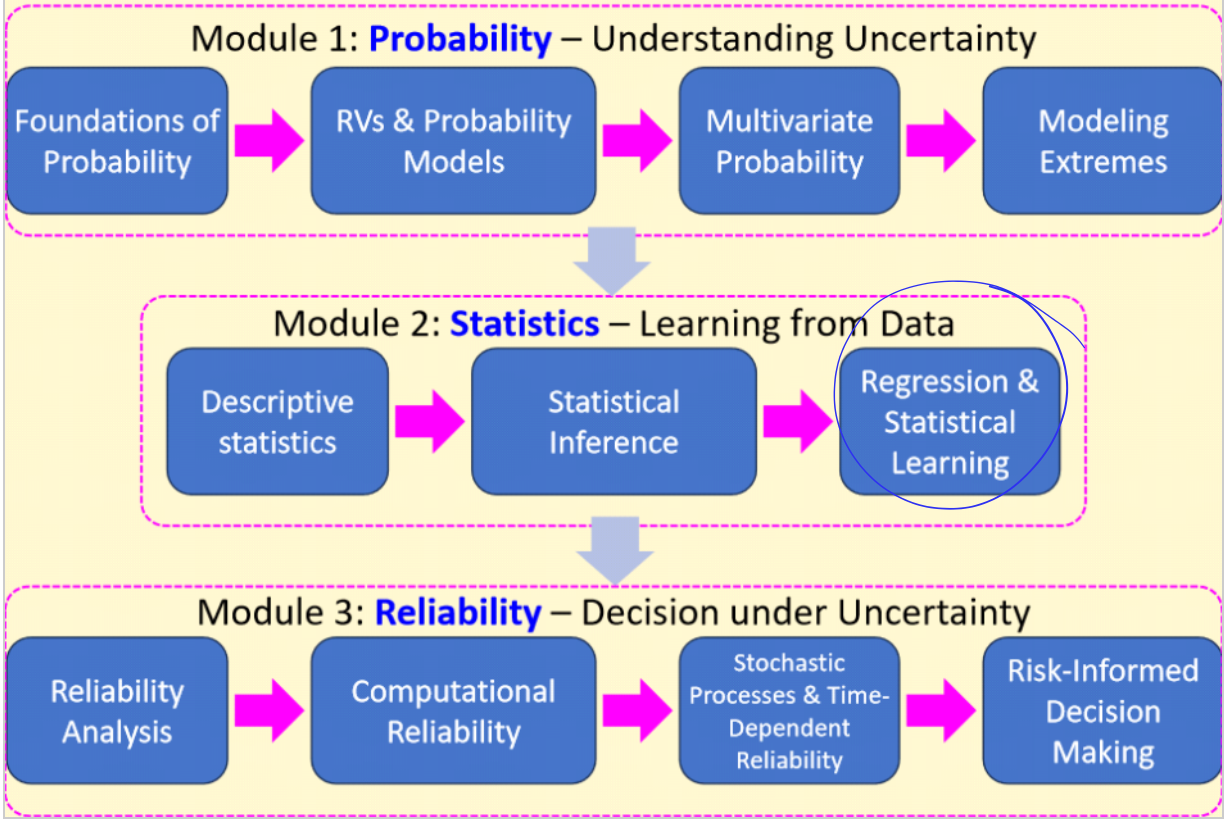
Dr. Prakash S Badal

Announcement

- Class/Tutorial rescheduling
  - No Class on 19th Oct → 23rd Oct
  - No tutorial on 19th Oct → Thu (?) 22nd Oct 10 - 11 am

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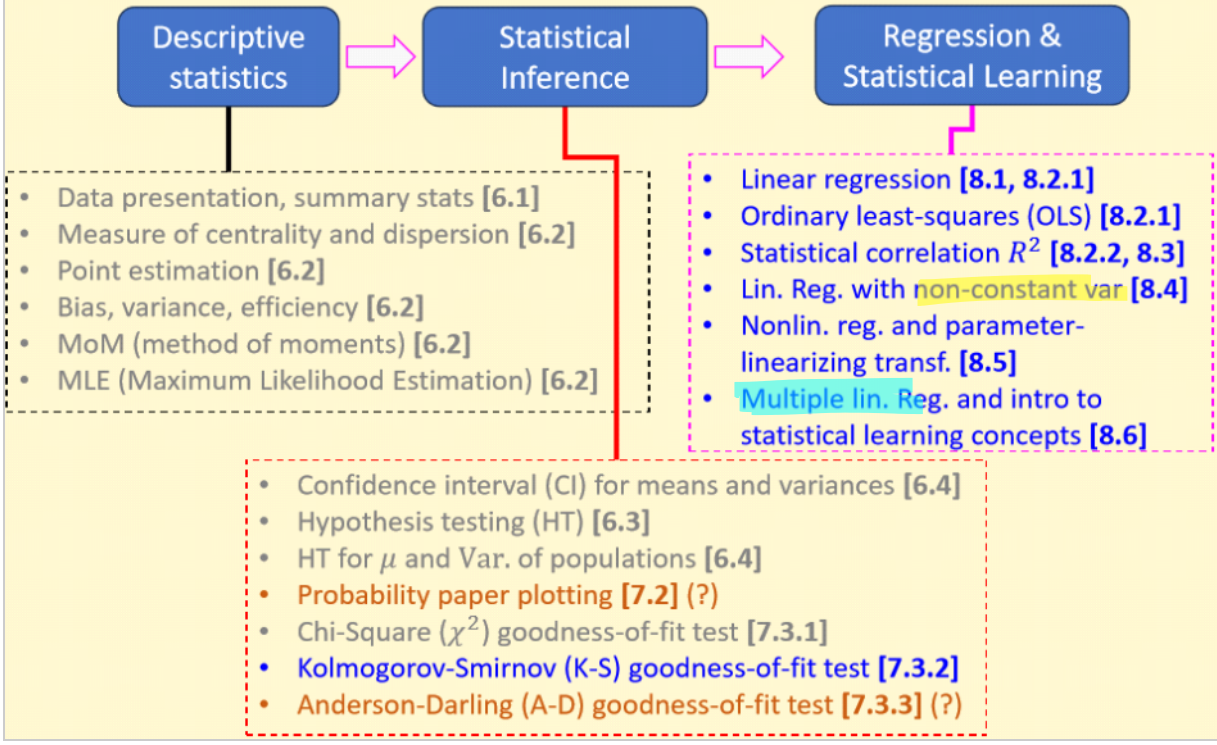
Course Flow



Regression & Statistical Learning (M2C3)

Module II: Chapter 3 (M2C3)

Module 2: **Statistics** – Learning from Data



M2C3

Ordinary Least Squares (OLS) Linear Regression: sum of squared errors, SSE $= \sum e_i^2$
Correlation & Model Evaluation: sample corr. Coeff. and R^2 to assess GoF.
Heteroscedasticity & Transformations: Weighted Least Squares (WLS) for non-constant Var. and parameter-linearizing transf. (e.g., Box-Cox) for nonlinear trends.
Multiple Linear Regression: Extend OLS to k-variable linear systems using matrix formulations ($\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$)
Surrogate Modeling & Probabilistic Machine Learning: Implement Gaussian Process Regression (Kriging) for rapid probabilistic surrogate modeling with spatial error bounds under uncertainty.

Motivation

How to map empirical relationships between variables to build predictive equations?

L-33 (1/3; Reg.)
Oct 13, 2026

Linear Regression

Ang and Tang (Section 8.1, 8.2.1)

Linear Regression and OLS

Regression of Y on X

$$E[Y \mid X = x'] = \alpha + \beta x'$$

α, β : constant

α : Intercept, β : Slope $\Rightarrow$ ALWAYS OBTAINED FROM DATA

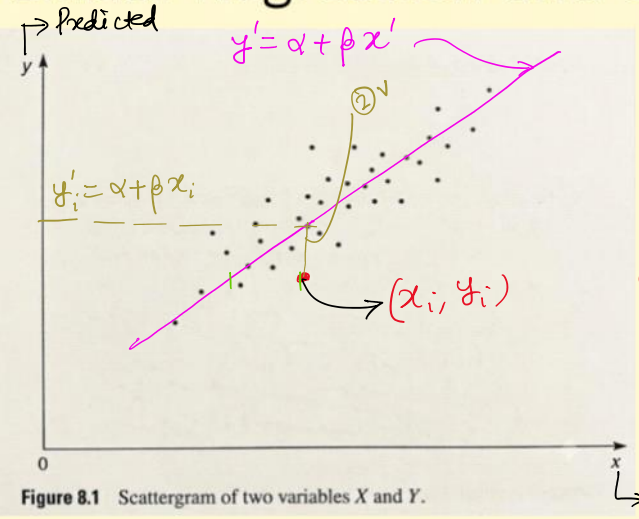


Figure 8.1 Scattergram of two variables X and Y.

Q) Which line "Best" Representations the data?

→ The one that minimizes the total "error"

$$\text{Error } \Delta = |y_i' - y_i| = |\alpha + \beta x_i - y_i| = |y_i - (\alpha + \beta x_i)|$$

Minimize Total Squared error

$$\Delta^2 = \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2$$

$$\frac{\partial \Delta^2}{\partial \alpha} = \sum -2(y_i - \alpha - \beta x_i) = 0 \Rightarrow n\bar{y} - n\alpha - n\beta\bar{x} = 0 \quad \times(-\bar{x})$$

$$\frac{\partial \Delta^2}{\partial \beta} = \sum -2(y_i - \alpha - \beta x_i)\, x_i = 0 \Rightarrow \sum x_i y_i - \alpha n\bar{x} - \beta \sum x_i^2 = 0 \quad (+)$$

$\sum x_i = n\bar{x}$

$\sum y_i = n\bar{y}$

Linear Regression and OLS

Linear Regression and OLS

$$\beta = \frac{n\bar{x}\bar{y} - \sum x_i y_i}{n\bar{x}^2 - \sum x_i^2}$$

$$\Rightarrow \hat{\beta} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$$

$$= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\hat{\alpha} = \bar{y} - \beta\bar{x}$$

estimates of Regression parameters $\hat{\alpha}$ $\hat{\beta}$

Regression Model

$$\hat{y} = \hat{\alpha} + \hat{\beta} x$$

$$E[Y \mid X = x] \equiv E[Y \mid x]$$

Variance in regression analysis

$$s^2_{Y|x} = \frac{\sum (y_i - \hat{y})^2}{n-2}$$

two degrees of freedom are lost on account of "TWO parameters being estimated from the same data"

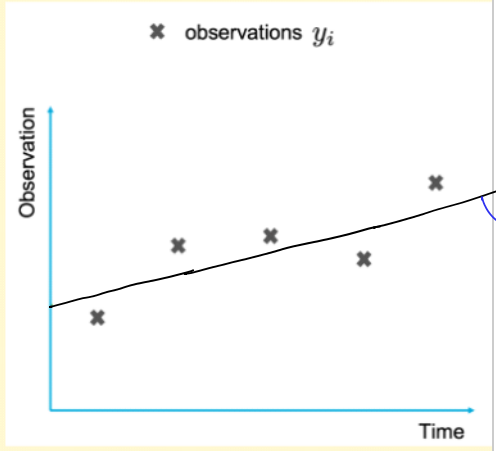
$$s^2_{Y|x} = \frac{\Delta^2}{n-2}$$

Least-square errors

A linear model with a single feature has two parameters:

An example:

$$y_1 = mx_1 + c$$
$$y_2 = mx_2 + c$$
$$\vdots$$
$$y_5 = mx_5 + c$$



$$\begin{Bmatrix} y_1 \\ y_2 \\ \vdots \\ y_5 \end{Bmatrix}_{5\times 1} = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_5 & 1 \end{bmatrix}_{5\times 2} \begin{Bmatrix} m \\ c \end{Bmatrix}_{2\times 1}$$

$$Y_{5\times 1} = X_{5\times 2}\, \beta_{2\times 1}$$

10 → 4

2 unknowns, 5 equations ⟹ NO UNIQUE SOLUTION

$$X^T_{2\times 5}\, Y_{5\times 1} = (X^T X)_{2\times 2}\, \beta_{2\times 1}$$

$$\Rightarrow \beta_{2\times 1} = (X^T X)^{-1} X^T Y$$ (MAGIC)?

Same as OLS estimator

Thank you!